# identity HTTP/TLS/JSON

## Interface Design Description

Service ID: *"identity"*

**Abstract**
This document describes the HTTP/1.1 protocol with TLS payload security and JSON payload encoding variant of the `identity` service of the Authentication system.

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# 1 Overview

This document describes the `identity` service interface of the Authentication system, which provides authentication of systems: a system logs in with its credentials and obtains an identity token, which it presents to other systems; providers verify such tokens.

It's implemented using protocol, encoding as stated in below table:

| Profile type | Type | Version |
|---|---|---|
| Transfer protocol | HTTP | 1.1 [1] |
| Data encryption | TLS | 1.3 [2], 1.2 [3] |
| Encoding | JSON | RFC 8259 [4] |
| Compression | N/A | - |
| Semantics | N/A | - |
| Ontology | N/A | - |

Table 1: Communication and semantics details used for the `identity` service interface

This document provides the Interface Design Description IDD to the *identity – Service Description* document [5]. For further details about how this service is meant to be used, please consult that document.

The rest of this document describes how to realize the `identity` service interface in detail. Both in terms of its operations (Section 2) and its information model (Section 3).

## 1.1 Interface registration

The service is registered in the ServiceRegistry with the interface template `generic_https` (`generic_http` when TLS is disabled), access port 8444 and base path `/authentication/identity`. The `operations` property of the interface lists the operations below with their method and path relative to the base path:

| Operation | Method | Path |
|---|---|---|
| `identity-login` | POST | `/login` |
| `identity-logout` | POST | `/logout` |
| `identity-change-credentials` | POST | `/change` |
| `identity-verify` | GET | `/verify` |

# 2 Service Operations

The interfaces of the `identity` service, its operations, data models and implementation are provided below. In particular, each subsection first names the HTTP method and path used to call the operation, after which it names the abstract operation from the `identity` SD document, as well as input and output types.

All operations require the requester's identity, presented according to the provider's authentication policy: the client certificate (mutual TLS) and, with the `outsourced` policy, `Authorization: Bearer IDENTITY-TOKEN//<token>`; systems holding the Authentication secret key send `Bearer AUTHENTICATOR-KEY//<system>//<hash>`, and with the `declared` policy `Bearer SYSTEM//<system>` is sent. Request and response bodies are `application/json`. Errors are answered with an ErrorMessage body and the status codes `400 Bad Request` (invalid parameter), `401 Unauthorized` (missing or invalid identity), `403 Forbidden` (not permitted, e.g. by the management or blacklist filter), `404 Not Found`, `408 Request Timeout`, `423 Locked`, `500 Internal Server Error` and `503 Service Unavailable` (a system it depends on failed).

## 2.1 POST /authentication/identity/login

**Operation:** identity-login
**Input:** IdentityRequest
**Output:** IdentityLoginResponse

Checks the credentials and opens a session. Returns the identity token and its expiry (`identity.token.duration`, 300 s in the default configuration). A previous token of the system becomes invalid.

A successful call returns `200 OK` with the output in the body.

```
POST /authentication/identity/login HTTP/1.1
Host: authentication:8444
Authorization: Bearer IDENTITY-TOKEN//<token>
Content-Type: application/json

{
  "systemName": "TemperatureSensor",
  "credentials": {
    "password": "<password>"
  }
}
```

Listing 1: An identity-login invocation.

```
{
  "token": "<token>",
  "expirationTime": "2026-10-22T12:00:00Z"
}
```

Listing 2: An identity-login response.

## 2.2 POST /authentication/identity/logout

**Operation:** identity-logout
**Input:** IdentityRequest

Checks the credentials and closes the session of the system.

A successful call returns `200 OK`.

```
POST /authentication/identity/logout HTTP/1.1
Host: authentication:8444
Authorization: Bearer IDENTITY-TOKEN//<token>
Content-Type: application/json
```

```
5
6  {
7    "systemName": "TemperatureSensor",
8    "credentials": {
9      "password": "<password>"
10   }
11 }
```

Listing 3: An identity-logout invocation.

## 2.3 POST /authentication/identity/change

**Operation:** identity-change-credentials
**Input:** IdentityChangeRequest

Checks the current credentials and replaces them with the new ones.
A successful call returns `200 OK`.

```
1  POST /authentication/identity/change HTTP/1.1
2  Host: authentication:8444
3  Authorization: Bearer IDENTITY-TOKEN//<token>
4  Content-Type: application/json
5
6  {
7    "systemName": "TemperatureSensor",
8    "credentials": {
9      "password": "<password>"
10   },
11   "newCredentials": {
12     "password": "<password>"
13   }
14 }
```

Listing 4: An identity-change-credentials invocation.

## 2.4 GET /authentication/identity/verify/*{token}*

**Operation:** identity-verify
**Input:** String
**Output:** IdentityVerifyResponse

Checks an identity token. Returns whether it is valid and, if so, the system, its sysop flag, login and expiry time. The caller must itself present a valid identity token (only `login`, `logout` and `change` are token-free).
Path parameter: *token*.
A successful call returns `200 OK` with the output in the body.

```
1  GET /authentication/identity/verify/<token> HTTP/1.1
2  Host: authentication:8444
3  Authorization: Bearer IDENTITY-TOKEN//<token>
```

Listing 5: An identity-verify invocation.

```
1  {
2    "verified": true,
3    "systemName": "TemperatureSensor",
4    "sysop": false,
5    "loginTime": "2026-10-22T12:00:00Z",
6    "expirationTime": "2026-10-22T12:00:00Z"
7  }
```

Listing 6: An identity-verify response.

# 3 Data Models

Here, all data objects that can be part of the service calls associated with this service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is meant to denote a JSON Object that must contain certain fields, or names, with values conforming to explicitly named types. Fields that are not needed may be omitted or `null` in requests; empty fields are omitted in responses. As a complement to the primary types defined in this section, there is also a list of secondary types in Section 3.6, which are used to represent things like hashes, identifiers and texts.

## 3.1 struct ErrorMessage

| Object Field | Value Type | Description |
|---|---|---|
| "errorMessage" | String | |
| "errorCode" | Number | |
| "exceptionType" | ExceptionType | |
| "origin" | String | |

## 3.2 struct IdentityChangeRequest

Changes the credentials of the requesting system.

| Object Field | Value Type | Description |
|---|---|---|
| "systemName" | String | Name of the system, PascalCase (e.g. `TemperatureSensor`). |
| "credentials" | Object<String> | Credentials of the authentication method, for `PASSWORD` the key `password`. |
| "newCredentials" | Object<String> | The new credentials. |

## 3.3 struct IdentityLoginResponse

The identity token issued at login.

| Object Field | Value Type | Description |
|---|---|---|
| "token" | String | The identity token, used as `Bearer IDENTITY-TOKEN//<token>`. |
| "expirationTime" | DateTime | UTC time when the identity token expires. |

## 3.4 struct IdentityRequest

System name and credentials.

| Object Field | Value Type | Description |
|---|---|---|
| "systemName" | String | Name of the system, PascalCase (e.g. `TemperatureSensor`). |
| "credentials" | Object<String> | Credentials of the authentication method, for `PASSWORD` the key `password`. |

## 3.5 struct IdentityVerifyResponse

Result of verifying an identity token.

| Object Field | Value Type | Description |
|---|---|---|
| "verified" | Boolean | True when the token is valid. |
| "systemName" | String | System the token belongs to. |
| "sysop" | Boolean | Whether that system is a system operator. |
| "loginTime" | DateTime | UTC time of the login. |
| "expirationTime" | DateTime | UTC time when the identity token expires. |

## 3.6 Primitives

As all messages are encoded using the JSON format [4], the following primitive constructs, part of that standard, become available. Note that the official standard is defined in terms of parsing rules, while this list only concerns syntactic information. Furthermore, the Object and Array types are given optional generic type parameters, which are used in this document to signify when pair values or elements are expected to conform to certain types.

| Type | Description |
|---|---|
| Boolean | JSON Boolean, `true` or `false`. |
| DateTime | JSON String with an ISO 8601 UTC timestamp, e.g. `2026-10-22T12:00:00Z`. |
| ExceptionType | One of `ARROWHEAD`, `INVALID_PARAMETER`, `AUTH`, `FORBIDDEN`, `DATA_NOT_FOUND`, `TIMEOUT`, `LOCKED`, `INTERNAL_SERVER_ERROR`, `EXTERNAL_SERVER_ERROR` (JSON String). |
| Number | JSON Number, an integer. |
| Object | JSON Object; Object<A> has string keys and values of type A. |
| String | JSON String. |

# 4 References

[1] R. Fielding, M. Nottingham, and J. Reschke, “Http/1.1,” IETF, RFC 9112, 2022.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “identity – Service Description (Authentication),” Eclipse Arrowhead, SD 5.2.2, 2026, authentication-v5.2.2-identity-SD.pdf.

# 5 Revision History

## 5.1 Amendments

| No. | Date | Version | Subject of Amendments | Author |
|---|---|---|---|---|
| 1 | 2026-10-22 | 5.2.2 | Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace. | Jerker Delsing |

## 5.2 Quality Assurance

| No. | Date | Version | Approved by |
|---|---|---|---|
| 1 | | 5.2.2 | |